A Multispectral Denoising Framework for Seismic Random Noise Attenuation

Yi Lin and Jinhai Zhang

***Abstract*— Random noise attenuation plays an important role in the seismic data processing. The global coherency among different spectral segments is often neglected in the traditional denoising methods, even though the seismic data are naturally broadband in the frequency spectrum. We proposed a multispectral denoising framework (MDF) for seismic random noise attenuation. The MDF contains three key components. First, we use a series of narrow-band filters on the noisy data to construct the multispectral data. Second, we normalize the constructed multispectral data to mitigate the amplitude discrepancy between different spectral slices. Third, we apply the intrinsic tensor sparsity regularization method to denoise the multispectral data. Numerical experiments on synthetic and field data show that the proposed framework can reduce the random noise as well as the erratic noise, and it can achieve improved denoised results, compared with a traditional method (i.e., the *f*-*x* deconvolution method) and the single-spectral version of the proposed method (i.e., the weighted nuclear norm minimization (WNNM) method). The proposed framework can be applied to both prestack and stacked data and generally well preserve the amplitude of useful signals, compared with the other two methods.**

***Index Terms*— 3-D tensor, low rank, multispectral denoising framework (MDF), random noise attenuation, seismic data processing.**

I. Introduction

RANDOM noise suppression is a crucial step in seismic data processing for oil and gas exploration. It can influence many subsequent processing steps [1], such as migration, waveform inversion, and interpretation. After decades of endeavor, a number of methods have been proposed to suppress the random noise, according to the differences between the signal and random noise.

One of the differences between the random noise and signal is that the random noise generally has almost no spatial coherency; in contrast, the signal is usually predictable along a spatial direction; thus, prediction-based methods [2]–[6] are proposed to denoise the data. However, these methods may be difficult to well distinguish the signal and random noise when their amplitudes are similar. One can solve this problem in a transformed domain, where the signal has high-amplitude coefficients (relatively compact in the transformed domain) while the random noise has small-amplitude coefficients (relatively widespread in the transformed domain). Therefore, the random noise can be attenuated by applying a simple thresholding operator over the amplitude of the coefficients in the transformed domain [7]. There are several candidates of the sparse transform: Fourier transform, Radon transform [8], curvelet transform [9]–[11], seislet transform [12], and shearlet transform [13], [14]. The above transforms

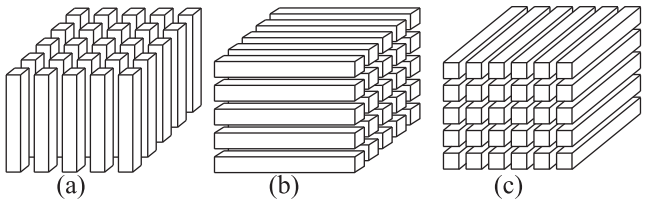


Fig. 1. Illustration of fibers of a third-order tensor [41]. (a) Mode-1 fibers. (b) Mode-2 fibers. (c) Mode-3 fibers.

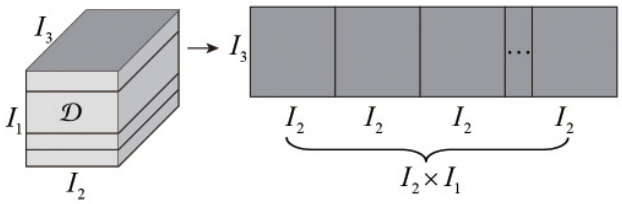


Fig. 2. Mode-1 unfolding of a third-order tensor $\mathcal{D}$ of dimensions $I_1 \times I_2 \times I_3$ [22].

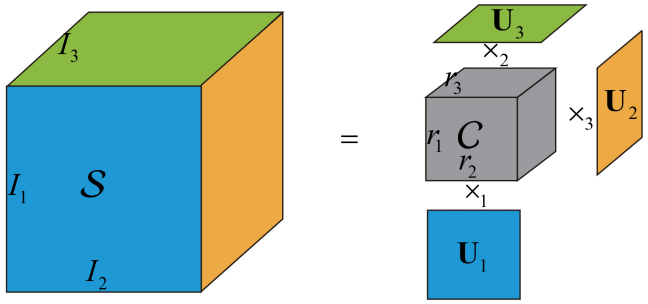


Fig. 3. Illustration of the Tucker decomposition of a three-order tensor [43].

Manuscript received October 9, 2020; revised November 19, 2020; accepted December 22, 2020. Date of publication January 14, 2021; date of current version December 3, 2021. This work was supported by the Strategic Priority Research Program of Chinese Academy of Sciences under Grant XDA17010404 and Grant XDA15011700 and in part by the National Natural Science Foundation of China under Grant 41941002. The work of Yi Lin was supported by the 68th China Postdoctoral Science Foundation under Grant 2020M680205. *(Corresponding author: Jinhai Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2020.3047633

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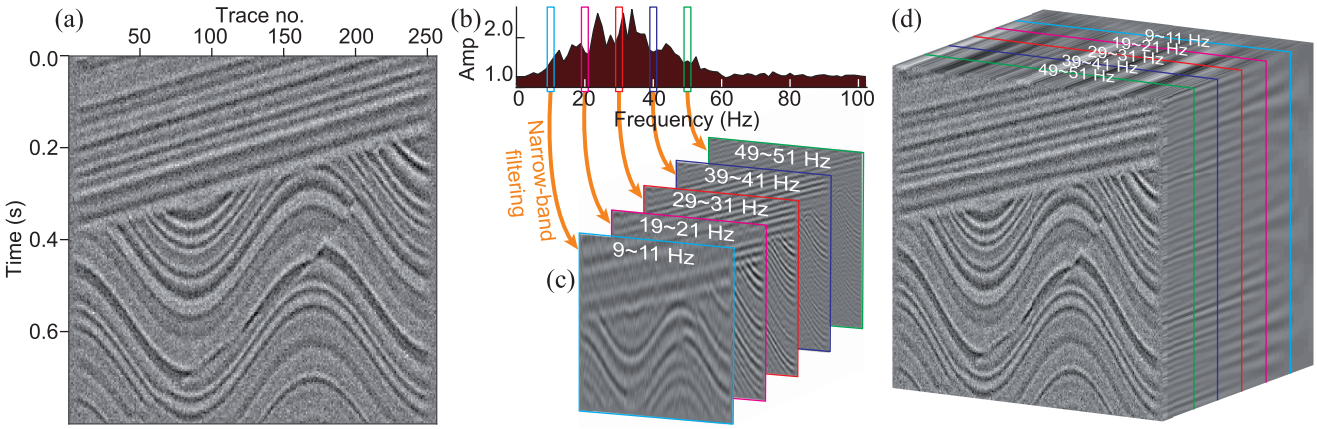


Fig. 4. Illustration of the procedure for constructing multispectral data. (a) Noisy data. (b) Amplitude spectrum of all the traces in (a). (c) Filtered results using the bandpass filter with five different ranges of the frequency band [colorful boxes in (b)]. (d) Final constructed multispectral data.

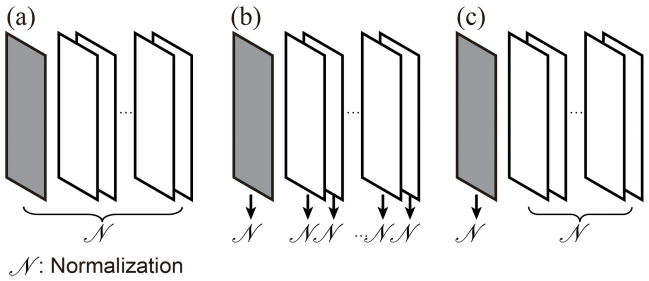


Fig. 5. Illustration of three types of normalization method. (a) Normalization-1. (b) Normalization-2. (c) Normalization-3. The grey box denotes the original noisy data slice.

have a known set of transform basis functions, and one can also learn them directly from the data to gain a sparser representation of the original data, which make it easier to separate the signal from the noisy data [15]–[17].

We can also separate the signal and random noise by some other characteristics, such as the low-rank property of the signal. The noise-free seismic data are of low-rank property under certain conditions (e.g., constructing a Hankel matrix from the original data [18]) while the noisy data can increase the rank; thus, we can attenuate the noise by reducing the rank. Rank-reduction-based methods include Cadzow filtering [19], multichannel singular spectrum analysis [18], and nuclear-norm minimization methods [20]–[22]. To further enhance the capability of denoising, the damped rank-reduction method [23], [24] is proposed, which can also be used for simultaneously denoising and interpolating the 3-D [25] and 5-D data [26]. The damped rank-reduction method was recently upgraded to the optimally damped rank reduction method [27] to deal with the rank-selection challenge.

Other types of denoising methods can also be utilized to suppress random noise. A typical type is the decomposition-based methods, which first decompose noisy data into constitutive components and then remove the one that represents noise. The decomposition-based methods include the singular value decomposition (SVD)-based method [28], empirical mode decomposition (EMD) [29], and continuous wavelet transform [30]. Another type is the time-frequency peak filtering method, which encodes the noisy seismic data as the instantaneous frequency of the analytic signal and then estimates the instantaneous frequency by detecting the peak of the time-frequency distribution to recover the signal of interest [31].

The repetitive characteristic of the seismic events has gained attention in the context of noise attenuation. Bonar and Sacchi [32] introduced the nonlocal means algorithm [33] from the field of digital image processing to denoise the seismic data. Then, Zhang and van der Baan [14] applied block matching [34] to group similar 2-D blocks into 3-D data arrays to enhance the coherent features, which improve the denoising performance of the traditional filtering process. Moreover, we note that the coherency can be further enhanced by investigating the similarity between the bandpass-filtered data using different filtering bands. Therefore, the denoising performance can be further boosted by incorporating the spectral similarity of the seismic data.

In this article, we propose a multispectral denoising framework (MDF), which is inspired by the intrinsic tensor sparsity regularization (ITSReg) [35] developed in the digital image processing field, to suppress the random noise in seismic data. The MDF is similar to the multiscale denoising method, which decomposes the seismic data into different spatial scales. Such multiscale denoising methods are often conducted by applying the empirical low-rank decomposition algorithm, e.g., the empirical dip filtering method [36], the empirical low-rank method [37], and the EMD-seislet method [38]. Instead, the MDF uses the similarity between different spectral slices, which can be viewed as a temporal scale and further boost the denoising performance. First, we construct the multispectral data by a series of narrow-band filters on the original noisy seismic data. Second, we normalize the multispectral data to balance the amplitude discrepancy between different spectral data slices. Finally, we apply the ITSReg method to denoise the multispectral data. Synthetic and field data examples demonstrated that the proposed MDF is suitable for both the prestack and stacked data sets and can better preserve the amplitude of useful signals, compared with the

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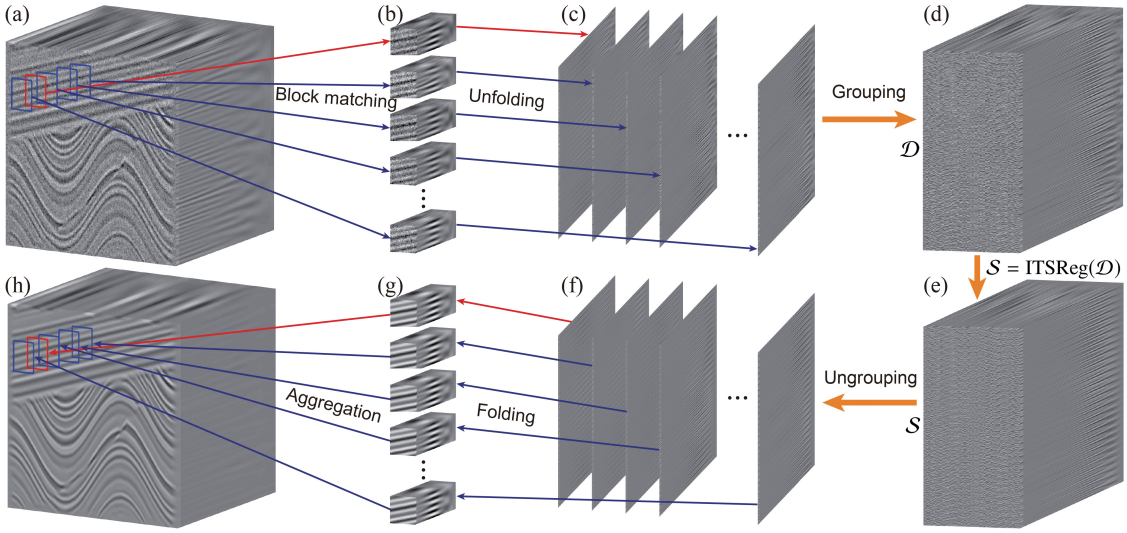


Fig. 6. Workflow of the MDF. (a) Multispectral noisy data tensor [Fig. 4(d)]. (b) Similar FBPs found by block matching. (c) Unfolded results of the FBPs. (d) 3-D tensor of the unfolded FBPs. (e) Results after applying the ITSReg method on (d). (f) Ungrouped results of the 3-D tensor (e). (g) Folded results of FBPs (f). (h) Final denoised result after aggregating the blocks (g).

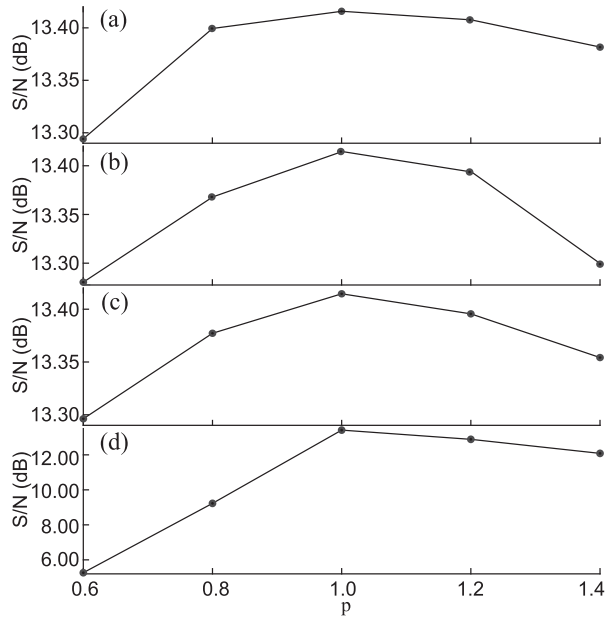


Fig. 7. Calculated S/N value for perturbation parameters. (a) $p\alpha$. (b) $p\delta$. (c) $p\mu$. (d) $p\rho$.

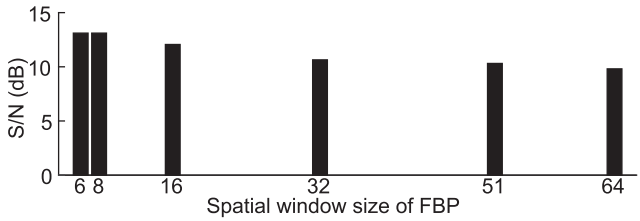


Fig. 8. Calculated S/N value for different spatial window sizes of FBP.

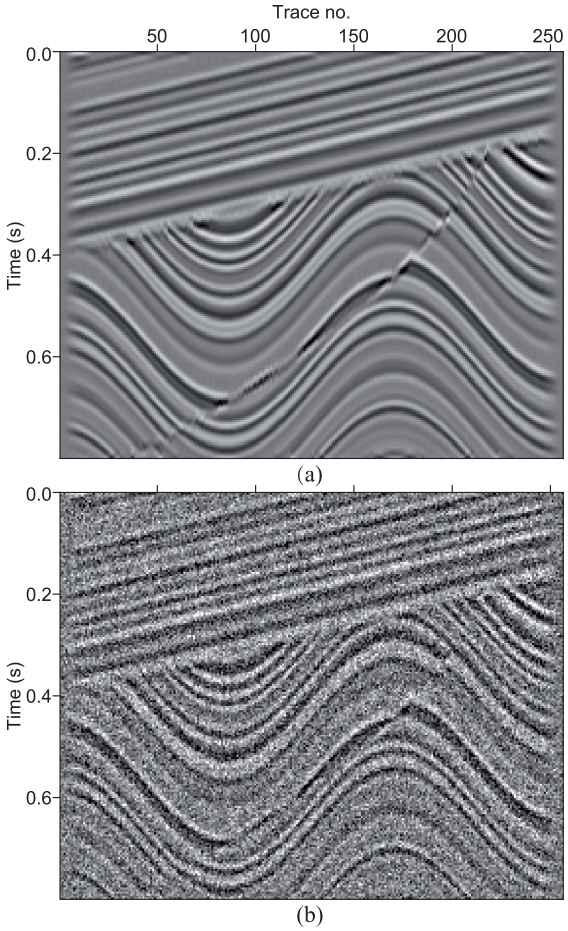


Fig. 9. Synthetic-1 seismic record. (a) Noise-free data. (b) Noisy data.

f-x deconvolution and the weighted nuclear norm minimization (WNNM) method [39], [40].

The rest of this article is organized as follows. In Section II, we present the workflow of the MDF. In Section III, we give the numerical examples and comparative discussions using synthetic and field data. In Section IV, we discuss the three factors that influence the performance, computational cost, and memory demand of MDF. Section V shows some conclusions.

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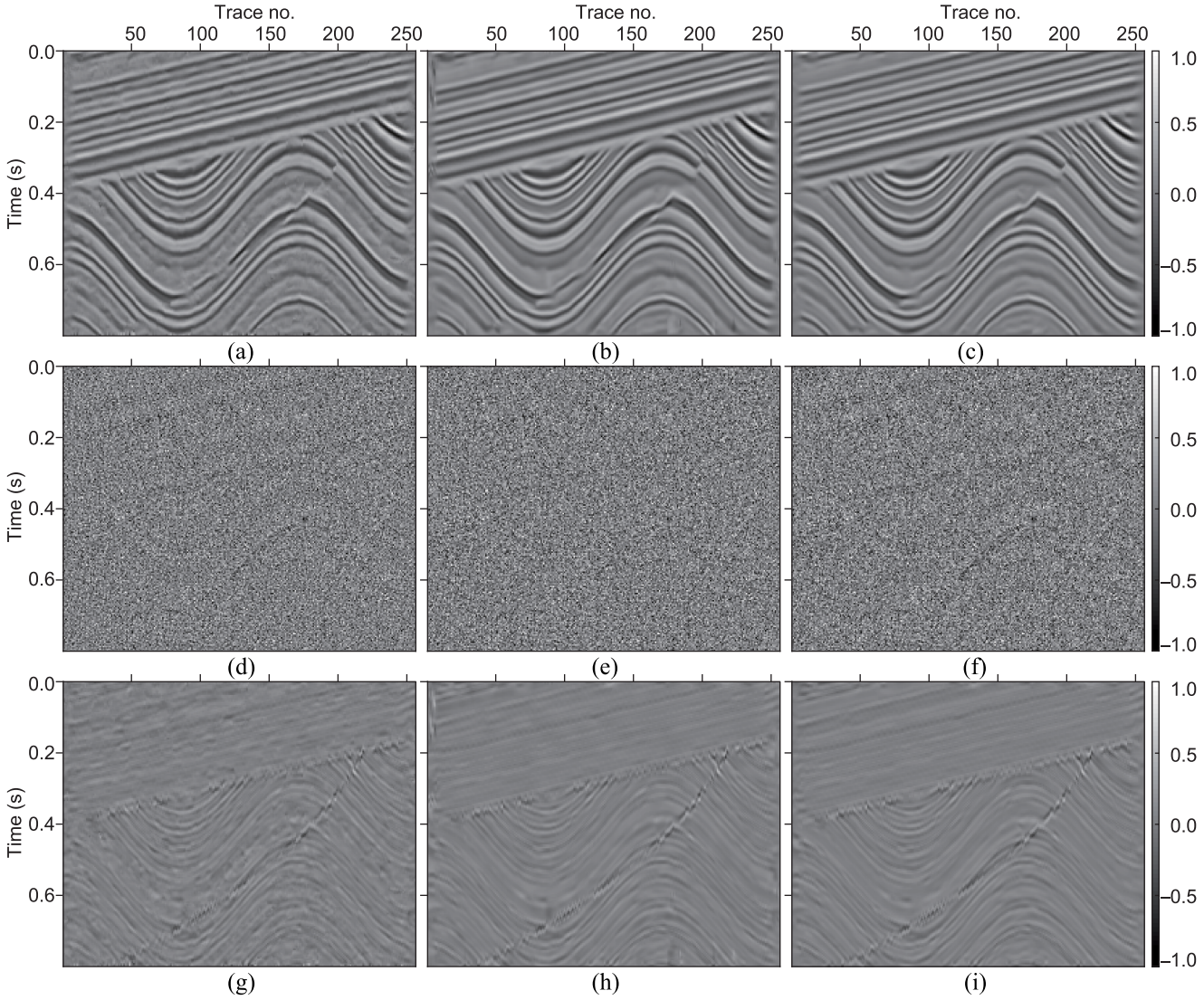


Fig. 10. Comparison of the denoised results between different normalization methods in the MDF. Denoised results are obtained by (a) Normalization-1, (b) Normalization-2, and (c) Normalization-3. (d)–(f) Removed noise sections obtained by (a)–(c), respectively. (g)–(i) Error panels caused by (a)–(c), respectively.

II. Theory

A. Notations

We denote scalars and vectors with lowercase italic letters (a) and lowercase boldface letters ($\mathbf{a}$), respectively. Matrices and tensors are denoted with uppercase boldface letters ($\mathbf{D}$) and calligraphic fonts $\mathcal{D}$, respectively. Elements of a multidimensional array $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_n}$ are denoted as their corresponding lowercase italic letters with subscripts ($x_{i_1 i_2 \cdots i_n}$).

Third-order tensors have the column, row, and tube fibers, which correspond to mode-1, mode-2, and mode-3 fibers, respectively, [41], as shown in Fig. 1. The mode-i unfolding of a tensor $\mathcal{D}$ (abbreviated as unfold$_i$ ($\mathcal{D}$) in the text) is a matrix symbolized by $\mathbf{D}_{(i)}$, which is obtained by considering the ith mode as the first dimension of a matrix and collapsing the other modes into the other dimension of that matrix. A third-order tensor can be unfolded in three different matrices. Fig. 2 illustrates the mode-1 unfolding of a third-order tensor $\mathcal{D}$. One can also unfold the tensor in the other two modes in a similar way. The mode-n folding of a matrix into a tensor (abbreviated as fold$_n(\cdot)$ in the text) is the inverse process of the mode-n unfolding of a tensor.

The mode-n product of a third-order tensor $\mathcal{D} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n} (n = 1, 2, 3)$ is denoted by $\mathcal{D} \times_n \mathbf{U}$ and is of size $I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_3$ [41], which takes the form as follows:

$$(\mathcal{D} \times_n \mathbf{U})_{i_1 \cdots i_{n-1} j i_{n+1} \cdots i_3} = \sum_{i_n=1}^{I_n} d_{i_1 i_2 i_3} u_{j i_n}. \tag{1}$$

The Tucker decomposition [42] of a third-order tensor $\mathcal{S} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is written as $\mathcal{S} = \mathcal{C} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_3$, where $\mathcal{C} \in \mathbb{R}^{r_1 \times r_2 \times r_3}$ is called the core tensor, and $\mathbf{U}_i \in \mathbb{R}^{I_i \times r_i}$ $(1 \leq i \leq 3)$ is composed of the r_i orthogonal bases along the ith mode of $\mathcal{S}$. Fig. 3 illustrates this decomposition.

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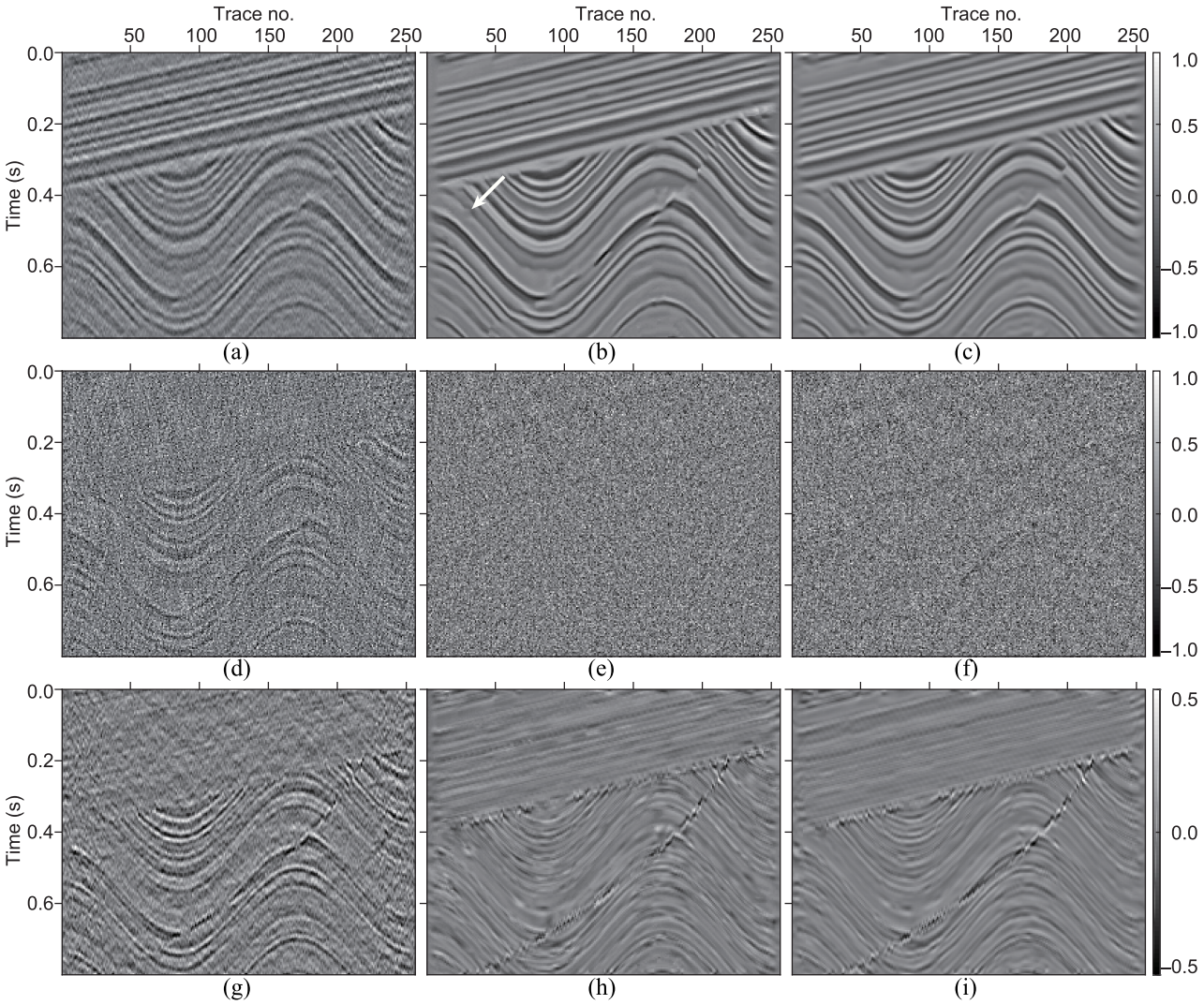


Fig. 11. Comparison of denoised results between different methods for the synthetic-1 data. The denoised result obtained by (a) f-x deconvolution method, (b) WNNM method, and (c) MDF. (d)–(f) Removed noise sections obtained by (a)–(c), respectively. (g)–(i) Error panels caused by (a)–(c), respectively.

B. Constructing Multispectral Data

The first step of the proposed MDF is to construct a multispectral data set, which can be generated by applying a series of narrow-band filters on the noisy data since the seismic data are naturally composed of multifrequency components. The coherency of the seismic data is enhanced after constructing such a multispectral data set. Fig. 4 illustrates this procedure. Fig. 4(a) shows a synthetic noisy seismic data set and Fig. 4(b) shows its amplitude spectrum in the frequency domain. A series of bandpass Butterworth filters are used to generate the multispectral data. The bandwidth of the Butterworth filter is chosen as follows: first, we plot the amplitude spectrum of the noisy data; then, we only use the dominant frequency components (0–60 Hz in this case) and divide them into many subband slices (e.g., 30 slices in this case), as shown in Fig. 4(c) (only a few slices are plotted). The bandwidth of the Butterworth filter is 2 Hz for this example. Note that these 2 Hz bandwidth filters do not overlap in order to obtain orthogonality between different subband spectra, which is critical for the performance of multispectral denoising. Empirically we find that 30 subband slices are enough to gain a good denoised result and this is the reason why we choose a bandwidth of 2 Hz to filter the data. From Fig. 4(c), we can observe the structural similarity between different subband slices. Fig. 4(d) shows the constructed multispectral data as a 3-D tensor, where the original noisy data are shown at the first frontal slice and the remaining slices are located according to their frequency bands.

C. Normalization

After applying the Butterworth bandpass filter to the original data, the amplitude of different subband slices may vary and will be different from the original noisy data. Therefore, the second step of the MDF is to normalize the multispectral data before applying any denoising methods. We have tested three normalization methods: normalizing the amplitude of whole multispectral data to [0, 1] [Fig. 5(a)], normalizing the amplitude of each slice of multispectral data to [0, 1]

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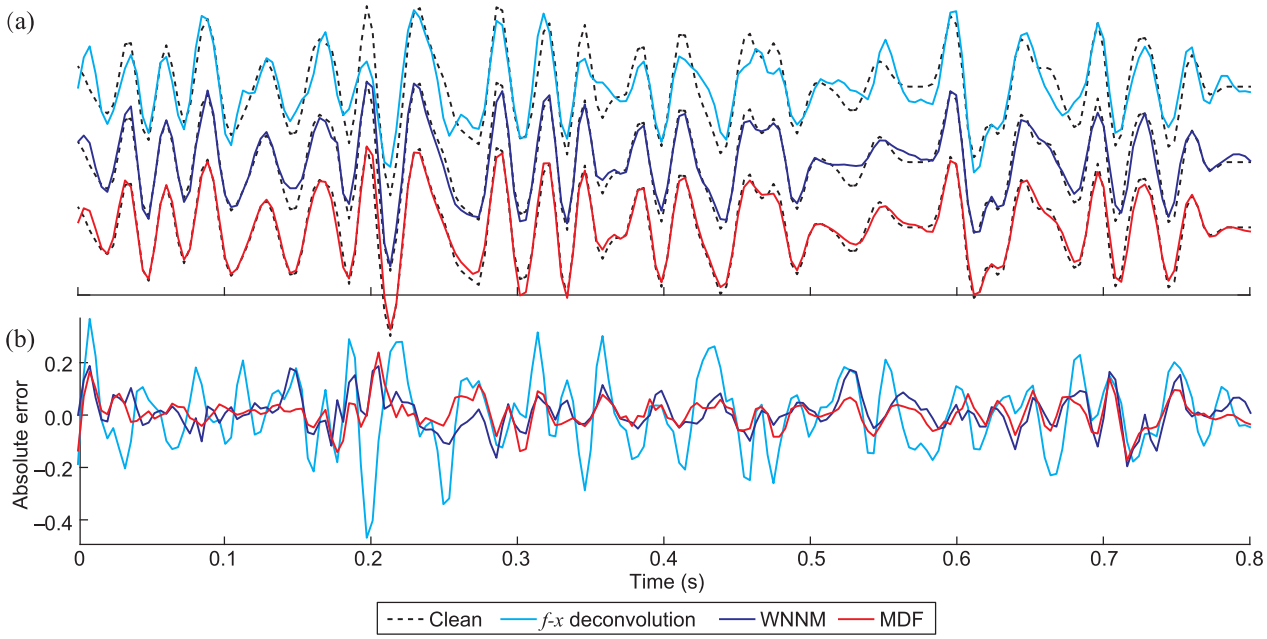


Fig. 12. Comparison of the denoised results (a) and residual errors (b) caused by different methods in terms of the waveform. The location is at trace no. 233 in Fig. 11. The cyan, blue, and red curves denote the results obtained by the f-x deconvolution, WNNM, and MDF, respectively. The dashed line denotes the noise-free data, which is taken as the reference result.

individually [Fig. 5(b)]; normalizing the amplitude of the original noisy slice and all the other slices generated by narrow-band filters to [0, 1] separately [Fig. 5(c)], since the amplitudes of the original noisy data are much stronger than that of the other slices generated by narrow-band filters. For convenience, we denote these three types of normalization methods as Normalization-1, Normalization-2, and Normalization-3, respectively. The performance of these normalization approaches will be examined later in Section III.

D. Multispectral Denoising of Seismic Data

Fig. 6 shows the workflow of the MDF. Fig. 6(a) shows the 3-D tensor representation of the multispectral seismic data, which requires a normalization to balance the amplitudes among different spectrum slices. For a given local full band patch (FBP) [the red patch in Fig. 6(a)] of the multispectral data, many similar FBPs [blue patches in Fig. 6(a)] exist in the whole 3-D tensor. We can apply a block matching method [34], [44] to find these similar FBPs, as shown in Fig. 6(b). Then, each FBP is unfolded to form a subband slice [Fig. 6(c)], which can be viewed as fixing the temporal direction and reshape the other two spatial dimensions into a vectorized data, and these slices are further concatenated to form an FBP group [Fig. 6(d)]. Next, the ITSReg method [35] is used to denoise each FBP group individually, as shown in Fig. 6(e). Finally, the denoised FBP groups are transformed back to the original domain by aggregating the similar FBPs. We summarized the implementation details of the MDF method in Algorithm 1. In Algorithm 1, we use the MATLAB notation $\mathcal{D}(:,:,k)$ denoting the kth frontal slice of the multispectral tensor $\mathcal{D}$. The superscript i in $\mathcal{D}^{(i)}$ denotes the ith iteration. The configuration of parameters in Algorithm 1 is listed in Section II-E. The third key component of the MDF is to apply the ITSReg method [35] to suppress the random noise. We review the ITSReg method in Appendix.

Algorithm 1 Algorithm for the MDF

Input: Noisy seismic data $\mathbf{d}_n$
Output: Denoised data $\mathbf{s}$
Initialization: constant parameter δ, and iteration L
Construct multispectral data $\mathcal{D}^{(0)}$ using narrow-band filters and let $\mathcal{D}^{(0)}(:,:,1) = \mathbf{d}_n$ and $\mathcal{S}^{(0)} = \mathcal{D}^{(0)}$
for $l = 1$ to L **do**
  Calculate $\mathcal{D}^{(l)} = \mathcal{S}^{(l-1)} + \delta(\mathcal{D}^{(l-1)} - \mathcal{S}^{(l-1)})$
  Construct the set of similar FBP group $\{\mathcal{D}_i\}_{i=1}^{K}$
  for each FBP group $\mathcal{D}_i$ **do**
    $\mathcal{S}_i = \text{ITSReg}(\mathcal{D}_i)$ // See details in Algorithm 2
  end for
  Aggregate $\{\mathcal{S}_i\}_{i=1}^{K}$ to form the denoised data $\mathcal{S}^{(l)}$
end for
Choose the denoised data $\mathbf{s} = \mathcal{S}^{(L)}(:,:,1)$ as the output
return $\mathbf{s}$

E. Parameter Settings

Several parameters have been involved in the MDF. We need to carefully choose these parameters to achieve better denoising results. We have listed the important parameters in Algorithms 1 and 2 (in Appendix). We usually need less than 3 iterations (i.e., the parameter L in Algorithm 1) to obtain an acceptable result. The parameter K in Algorithm 1 is related to the patch size of the block matching algorithm. For example, if the seismic data are of dimension N_t by N_x, and the patch size of the block matching algorithm is n, the maximum K to be evaluated is $K = (N_t - n) \cdot (N_x - n)$. Usually, many similar patches exist in seismic data, which will greatly reduce the K value. Readers may refer to [34] to explore more details.

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We provide some sensitivity tests against different parameters here to illustrate how to choose the parameters appropriately. The signal-to-noise ratio (S/N) is taken as a criterion to measure the quality of random noise attenuation, which is defined as

$$S/N = 20\log_{10}\frac{\|\mathbf{D}_0\|_2^2}{\|\mathbf{D}_0 - \mathbf{D}\|_2^2} \tag{2}$$

where $\mathbf{D}$ is the noisy seismic data (or denoised seismic data), $\mathbf{D}_0$ is the noise-free seismic data, and $\|\cdot\|_2$ describes the L_2 norm.

We follow a strategy used in our previous study [45] to conduct the sensitivity experiment, which involves four parameters: ρ, μ (in Algorithm 2), δ (in Algorithm 1) and α. By imposing a perturbation value (say $p \in [0.6, 1.4]$) on the initial value of the parameters, we can study the relationship between the S/Ns of the denoised result and perturbing parameters. We use the sigmoid data [46] (called synthetic-1 in figures) with varying input noise level (from -8 to 0 dB) to study such relations. Fig. 7 shows the S/N of denoised data versus perturbing parameters of ρ, μ, δ, and α. The S/N change caused by perturbing parameters μ, δ, and α is less than 0.2 dB and the S/N of the denoised data is rather high, which suggests that these three parameters are very stable against different noise levels. We suggest to set μ, δ, and α to 250, 0.33, and 0.091, respectively, which can guarantee a reasonable denoising result. The S/N change caused by the perturbing parameter ρ is quite large, however, the peak S/N usually acquires when we set ρ to 1.2. Besides, we test the sensitivity of the MDF on different FBP window sizes (i.e., 6, 8, 16, 32, 51, 64), as shown in Fig. 8. It turns out that a smaller FBP window size (6–8) usually performs better than larger ones. Nevertheless, one can always carefully tune all these parameters to achieve better results for different seismic data sets.

III. Numerical Examples

We apply the MDF on two synthetic data sets and two field data sets, and compare its performance with a traditional method (i.e., the f-x deconvolution method [2]) and the single-spectral version of the proposed method (i.e., the WNNM method [39]. For the f-x deconvolution method, we use an implementa-tion from a widely used package named SeismicLab (http://seismic-lab.physics.ualberta.ca/). Two key parameters: the length of the filter l_f and a prewhitening para-meter μ_f, are provided in each example. For the WNNM method, we use the official implementation (http://www4.comp.polyu.edu.hk/čslzhang/code/WNNM_code.zip), only one key parameter δ_w (an estimate of the variance of the noise) is involved. For the MDF, the basic parameters are given in Section II-E, and the rest of the key parameters like the patch size K and subbandwidth used for constructing multispectral data w_m are given in each example.

A. Normalization Results

To illustrate the performance of different normalization approaches, we use the sigmoid data, which are created to mimic a premigrated section. The synthetic-1 data have 200 samples in time and 256 traces in space. The noise-free and noisy data (with Gaussian white noise) are shown in Fig. 9(a) and (b), respectively. The denoised results by the three types of normalization approaches are shown in Fig. 10(a)–(c), respectively, and the corresponding noise sections are shown in Fig. 10(d)–(f), respectively. We observe that Normalization-1 loses more useful signals than the other two methods since the amplitude of the multispectral slices varies but Normalization-1 acts on the whole data set. We also calculate the S/Ns of denoised results obtained by different types of normalization methods, as shown in Table I. Although the S/N of Normalization-2 has the highest value, the denoised result seems to have more artifacts than Normalization-3. This is because the amplitude of the original noisy data is much stronger than that of the other slices generated by narrow-band filters and Normalization-3 can better address this issue. Therefore, we adopt Normalization-3 in our later numerical examples.

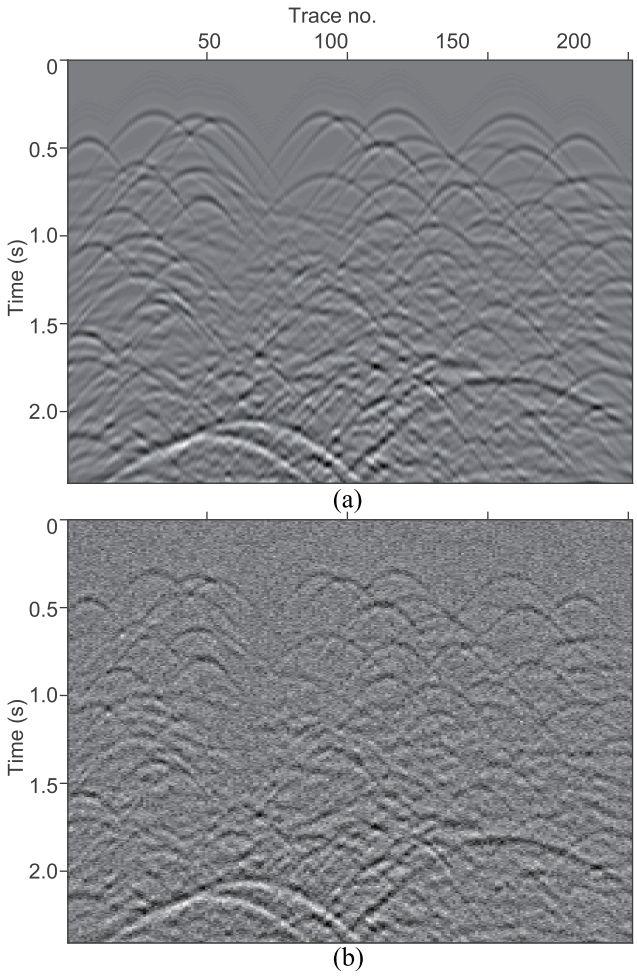


Fig. 13. Synthetic-2 seismic record. (a) Noise-free data. (b) Noisy data.

B. Synthetic Examples

We compare the denoising performance of three methods on the sigmoid data. Fig. 11(a)–(c) shows the denoised results using the f-x deconvolution, WNNM, and MDF, respectively. The S/Ns of denoised data are shown in the first row

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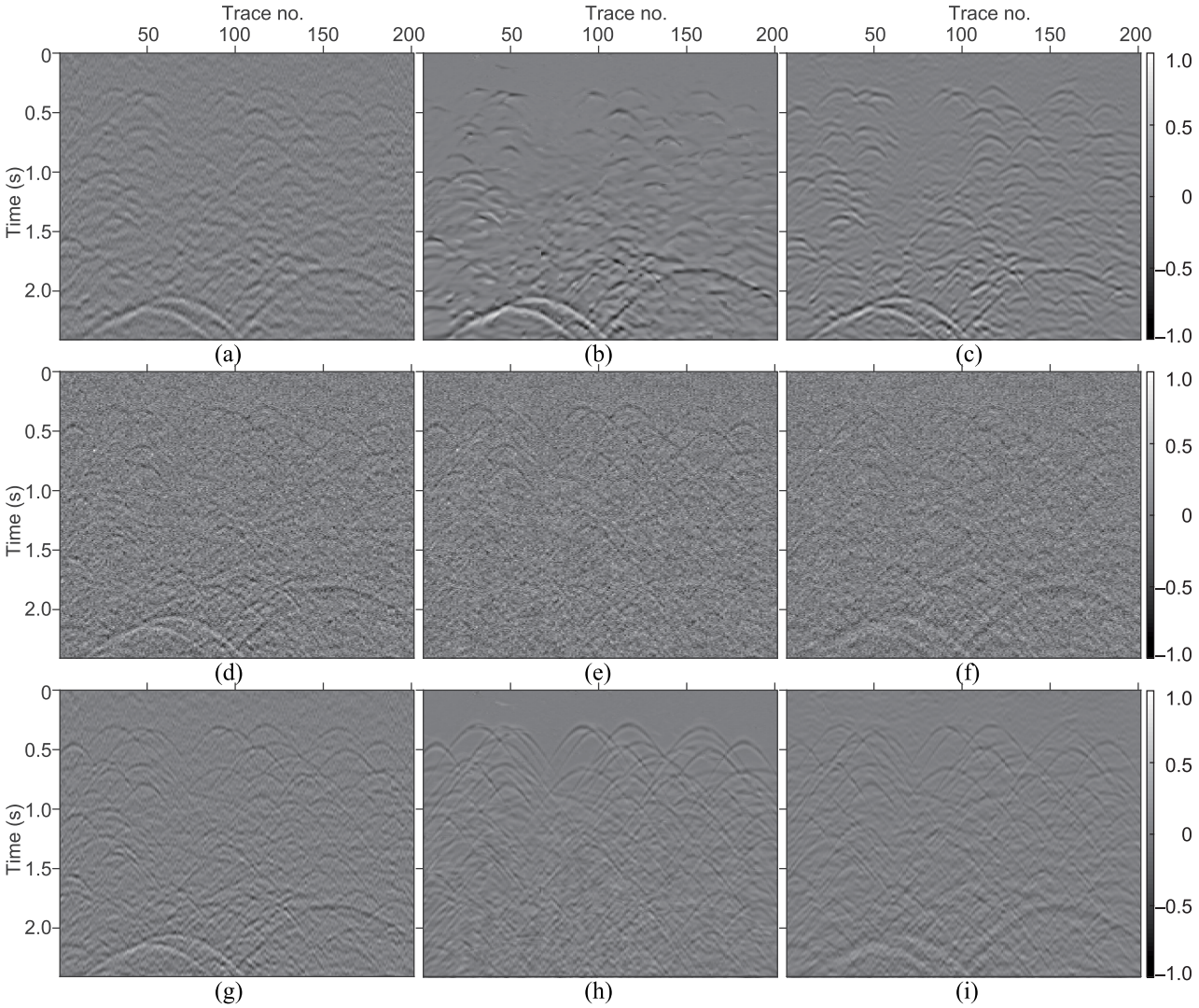


Fig. 14. Comparison of denoised results between different methods for the synthetic-2 data. The denoised result obtained by (a) f-x deconvolution method, (b) WNNM method, and (c) MDF. (d)–(f) Removed noise sections obtained by (a)–(c), respectively. (g)–(i) Error panels caused by (a)–(c), respectively.

TABLE I

COMPARISON OF S/N IN DECIBELS BETWEEN THE DENOISED RESULTS OBTAINED BY DIFFERENT NORMALIZATION IN THE MDF FOR THE SYNTHETIC-1 MODEL[1]

Model	Input	Normalization-1	Normalization-2	Normalization-3
Synthetic-1	−0.84	10.86	12.46	12.42

[1] Synthetic-1 data is shown in Fig. 9. The S/N value is calculated using (2).

TABLE II

COMPARISON OF S/N IN DECIBELS BETWEEN THE DENOISED RESULTS OBTAINED BY DIFFERENT METHODS[1]

Model	Input	f-x deconvolution	WNNM	MDF
Synthetic-1	−0.84	6.54	11.83	12.42
Synthetic-2	−4.00	3.47	5.21	5.69

[1] Synthetic-1 and -2 data are shown in Figs. 9 and 13, respectively. The S/N value is calculated using (2).

of Table II. The length of the filter l_f and a prewhitening μ_f for f-x deconvolution method are 15 and 0.0025, respectively. The estimate of the variance of the noise δ_w for the WNNM method is 0.3. The patch size K and subbandwidth used for constructing multispectral data w_m for the MDF are 7 and 2 Hz, respectively. The S/N of the MDF is the highest among all these three approaches listed. Among the noise sections shown in Fig. 11(d)–(f), the f-x deconvolution method leaks a large portion of useful signals; the WNNM method has some distortions in the denoised result [denoted by the white arrow in Fig. 11(b)]; in contrast, the MDF only has some weak low-frequency content left in the noise section. The error panels of these three methods are shown in Fig. 11(g)–(i). Again, the overall absolute error caused by the MDF is the smallest among these three methods. Therefore, the MDF well preserves the details of seismic events and leaks the least useful signal compared with the other two methods.

Fig. 12(a) compares the denoised results with noise-free data in terms of the waveform at trace no. 233. The cyan,

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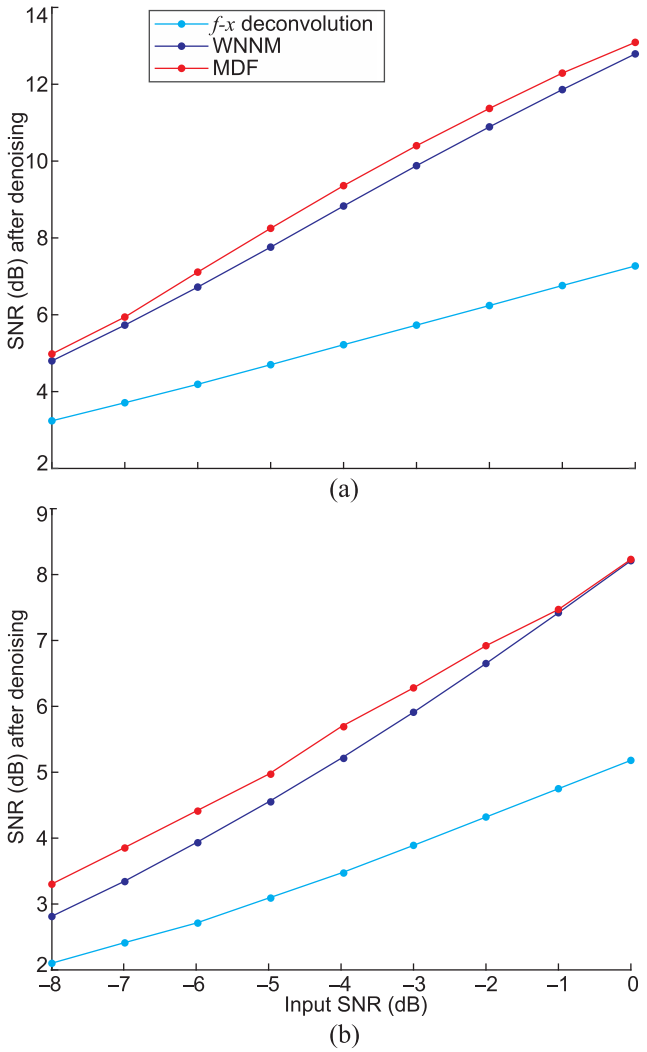


Fig. 15. Comparison of the denoising capability with different input noise levels. (a) Synthetic-1 model. (b) Synthetic-2 model.

blue, and red curves denote the results obtained by the f-x deconvolution method, the WNNM method, and MDF, respectively. The dashed line denotes the noise-free data, which is taken as reference. The denoised results obtained by the MDF are more consistent with the noise-free data than that of the other two methods. Fig. 12(b) gives the recovered error. The recovered error of the MDF is the lowest among these three methods, which indicates the good performance of the MDF in amplitude preservation.

Fig. 13(a) shows another synthetic data set (called synthetic-2 in the figures). The synthetic-2 data have 201 traces in space and 301 temporal samples in each trace, with a sampling interval of 0.008 s. Strong Gaussian white noise is added to noise-free data [Fig. 13(b)]. The length of the filter l_f and a prewhitening μ_f for f-x deconvolution method are 15 and 0.0025, respectively. The estimate of the variance of the noise δ_w for the WNNM method is 0.241. The patch size K and subbandwidth used for constructing multispectral data w_m for the MDF are 7 and 0.83 Hz, respectively. Fig. 14(a)–(c) shows the denoised results using the f-x deconvolution method, the WNNM method, and MDF, respectively. Fig. 14(d)–(f) shows the corresponding noise sections that are estimated from the noisy data. The denoised results by the f-x deconvolution method remain a lot of random noises, and the results by the WNNM method is oversmoothed with distorted events; in contrast, the results by the MDF are much cleaner and the amplitude is also stronger than that of the other two methods. Fig. 14(g)–(i) shows the error panels of these three methods. The error panel of the MDF has the least error in most locations, which indicates that the MDF can preserve the amplitude of desired signals while removing random noises. We also quantitatively assess the denoising performance via the S/N computed before and after the random noise attenuation. The S/Ns are shown in the second row of Table II. The S/N of the denoised data using the MDF is the highest, which confirms the effectiveness of the proposed framework.

C. Sensitivity to Different Noise Levels

We further test these three methods on two synthetic models with various noise levels. The input S/N of the noisy data varies from a low S/N value (-8.0 dB) to high (0 dB). The S/Ns of the denoised results obtained by the three methods for two models are shown in Fig. 15(a) and (b), respectively. We observe that the MDF can achieve the best denoising performance among these three methods for various noise levels.

D. Application to Erratic Noise Attenuation

Unlike random noise, the erratic noise [47], [48], which designates non-Gaussian noise that consists of large isolated events with known or unknown distribution, is usually difficult to be dealt with. We apply the MDF to an erratic plus random noisy data set, shown in Fig. 16(a), and compare its denoising performance with the other two methods. The clean data are the same as Fig. 9(a). The denoised results by the f-x deconvolution method, the WNNM method, and MDF are shown in Fig. 16(b)–(d), respectively. We observe some distortions of events in the denoised result obtained by the f-x deconvolution method. The denoised result by the WNNM method is much cleaner, but there are some erratic noise remained in the denoised section. The MDF can simultaneously remove erratic noise as well as random noise, and it gains a result that is close to the noise-free data. The noise section obtained by the MDF [Fig. 16(h)] has the least signal leakage compared with the other two methods [Fig. 16(f)–(g)], which again verifies that the MDF can achieve the best denoising performance among these three methods even for erratic noisy data sets.

E. Field Data Application

We further test these three denoising methods on two field seismic data sets. The first one (called field-1 data) was downloaded from the United States Geological Survey (USGS) Energy Resources Program website (https://certmapper.cr.usgs.gov/data/apps/npra/, accessed 22 September 2020). Fig. 17 shows a portion of a stacked data set (line ID 1402). The field-1 data have a time sampling interval of 0.008 s and have 380 traces. Random and coherence noise coexist in this example. The length of the filter l_f and a prewhitening μ_f for f-x deconvolution method are 3

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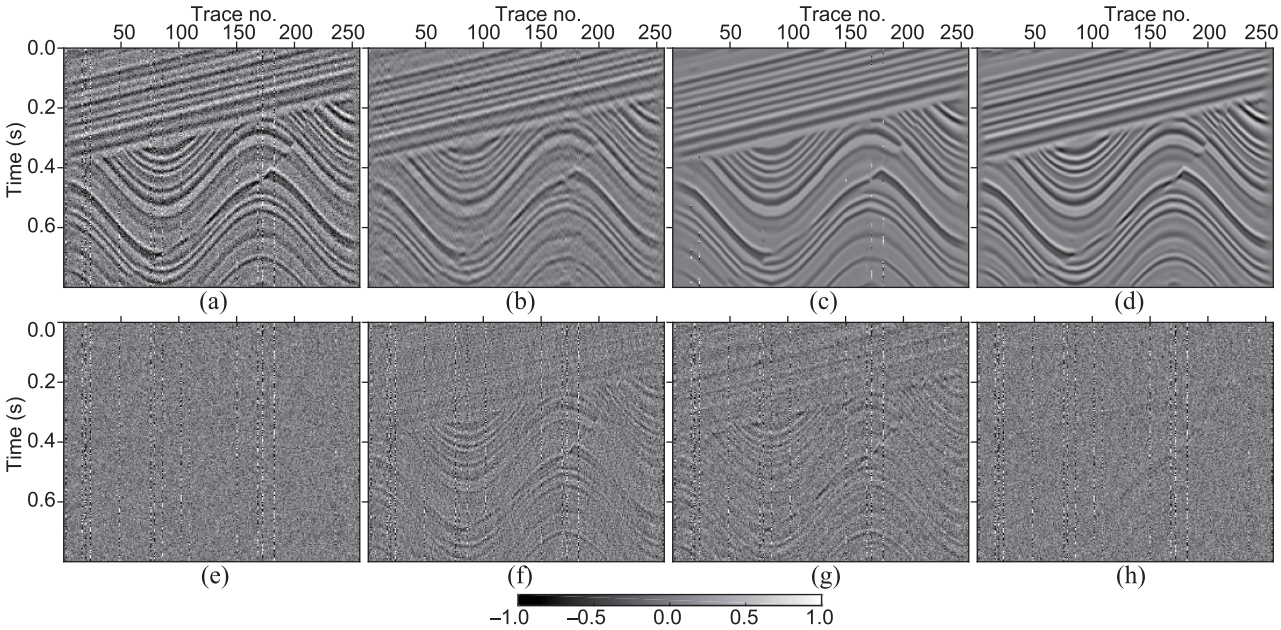


Fig. 16. Comparison of the denoised result obtained by different methods for the synthetic-1 data with added erratic plus random noise. (a) Erratic plus random noisy data. The denoised result obtained by (b) f-x deconvolution method, (c) WNNM method, and (d) MDF. (e) Added erratic plus random noise. The noise sections obtained by (f) f-x deconvolution method, (g) WNNM method, and (h) MDF.

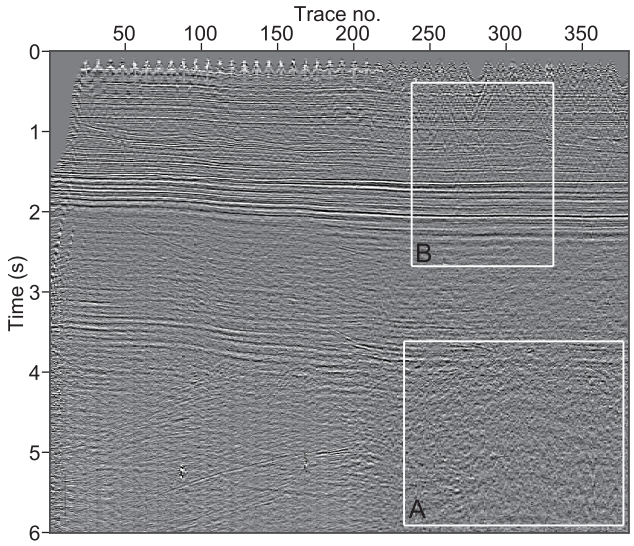


Fig. 17. Noisy field-1 data. The white boxes denote two enlarged areas for detailed comparison.

and 0.00125, respectively. The estimate of the variance of the noise δ_w for the WNNM method is 0.013. The patch size K and subbandwidth used for constructing multispectral data w_m for the MDF are 6 and 2 Hz, respectively. Fig. 18(a)–(c) shows the denoised results using these three methods, respectively. We observe that the denoised result of the WNNM method is again oversmoothed [Fig. 18(b)]. The corresponding noise sections are shown in Fig. 18(d)–(f), respectively. Many seismic events are removed by the f-x deconvolution method [Fig. 18(d)]; in contrast, only some weak low-frequency components of the horizontal events are removed by the MDF, as shown in Fig. 18(f).

We compare the amplitude spectrum of the denoised results in Fig. 19. The amplitude spectrum of the MDF is the highest among these three methods around the dominant frequency (from 15 to 45 Hz), which indicates that the MDF can better preserve the amplitude of useful signals. We then compare the waveform of the denoised result as well as the original field data at trace no. 100, 200, and 300, shown in Fig. 20. We observe that the MDF can better preserve the edge of the original data. We further show the enlarged view of the denoised results and their corresponding noise sections for two regions denoted by the white boxes in Fig. 17, as shown in Figs. 21 and 22. For the enlarged area A, some random noise is still visible in the denoised results by the f-x deconvolution method, as shown in Fig. 21(b); the denoised result by the WNNM method is oversmoothed, as shown in Fig. 21(c); in contrast, the amplitude of the events in the denoised result by the MDF is stronger than that of the other two methods [see the white arrows in Fig. 21(b)–(d)]. For the enlarged area B, we observe that a large portion of coherence noise remained in the denoised result by the WNNM method, as shown in the white box in Fig. 22(c); some energy of the horizontal events is removed by the f-x deconvolution method, as shown by the white arrows in Fig. 22(e); in contrast, the overall behavior of the MDF is better than that of the other two methods, which removes most of the linear coherence noise and random noise.

The second field data (called field-2 data) is a shot gather, downloaded from https://wiki.seg.org/wiki/2D_Vibroseis_Line_001#Download_Link. It has a time sampling interval of 0.004 s and has 282 traces, as shown in Fig. 23. We mute the direct wave and some surface wave first; then, we apply the three methods to denoise the data. The length of the filter l_f and a prewhitening μ_f for f-x deconvolution

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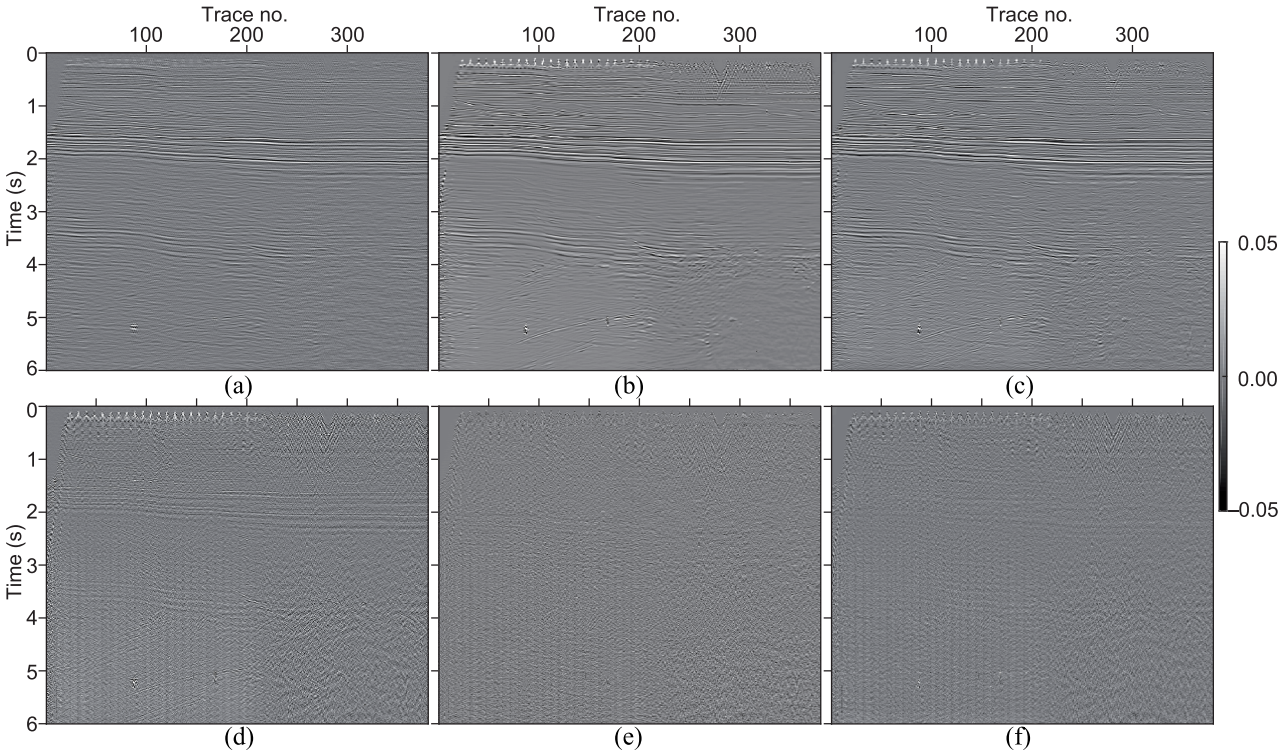


Fig. 18. Comparison of the denoised results and corresponding noise sections for the field-1 data. Denoised results are obtained by (a) f-x deconvolution method, (b) WNNM method, and (c) MDF, respectively. (d)–(f) Removed noise sections obtained by (a)–(c), respectively.

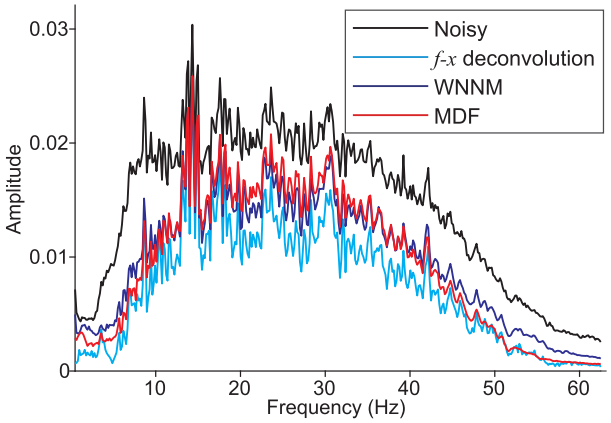


Fig. 19. Comparison of the amplitude spectrum between the field-1 data and the denoised results.

method are 15 and 0.00125, respectively. The estimate of the variance of the noise δ_w for the WNNM method is 0.15. The patch size K and subbandwidth used for constructing multispectral data w_m for the MDF are 6 and 3.17 Hz, respectively. The denoised results and their corresponding noise sections are shown in Fig. 24. The noise section of the f-x deconvolution method leaks some useful events, as denoted by the arrow in Fig. 24(d). The denoised results obtained by the WNNM method and MDF are better than that by the f-x deconvolution method; however, the amplitude of events by the WNNM method is weaker than that by the MDF, as denoted by the arrows in Fig. 24(b) and (c). The two field data sets show that the MDF can effectively reduce the random noise in both prestack and stacked profiles.

IV. Discussions

The performance of the MDF depends on three factors: the S/N of the input noisy data, the normalization of the multispectral seismic data, and the proper selection of frequency bands. We use 30 narrow-band filters from 0 to 60 Hz in synthetic-1 data, which corresponds to the effective bandwidth of the data. These narrow-band filters have some overlap in frequency with their neighbors. One can try higher-order Butterworth filters or other filters (say Chebyshev) to construct the multispectral data with less overlap. A smaller overlap among each group of adjacent narrow-band filters can provide a better separation of frequency components, which may help suppress noises with particular frequency components.

Normalization of the multispectral seismic data may influence the amplitude; thus, we try to recover the amplitude after denoising using the inverse operation of the normalization. Nevertheless, the amplitude may still have some deviation from the true amplitude, which may harm the amplitude-sensitive applications. Numerical experiments show that our method can better preserve the amplitude, compared with the other two methods. The very low-frequency components are easy to be harmed due to their too low amplitudes, as shown in Figs. 11(f) and 14(f). More complex normalization ways may help to mitigate this effect. Besides, one can use a post-process strategy to alleviate this issue by [49], since it can recover the visible lost components in the removed noise section.

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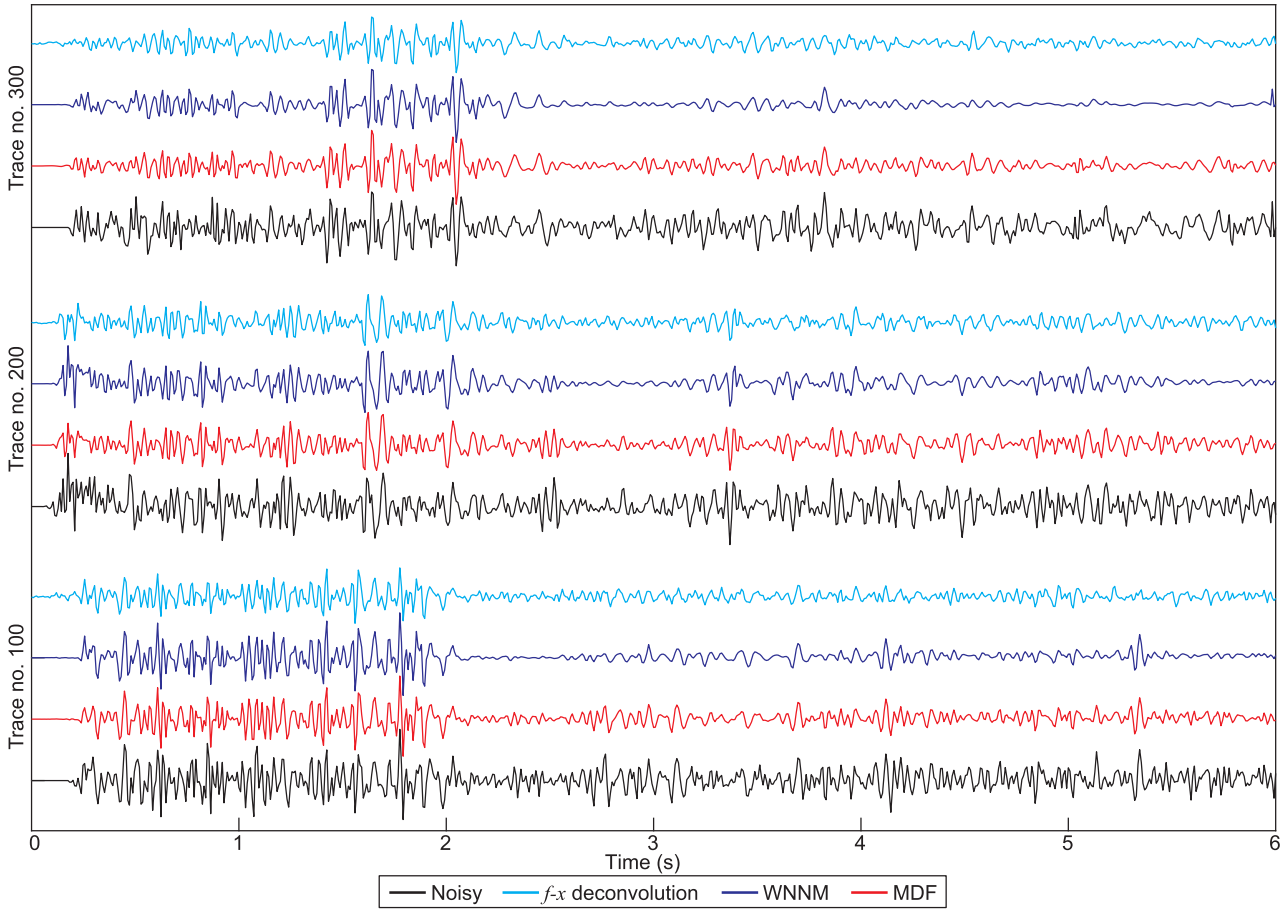


Fig. 20. Comparison of the denoised results obtained by different methods in waveform view for field-1 data. The location is at trace no. 100, 200, and 300. The cyan, blue and red curves denote the results obtained by the f-x deconvolution, WNNM, and MDF, respectively. The black line denotes the noisy field data.

Many traditional denoising methods may not perform well in the prestacked volumes but they can obtain a good result in stacked volumes since the assumptions vary in different data domains. Seismic data in prestack volumes often have dipper events as well as hyperbolic events whose dip angle changes corresponding to different spatial locations. Therefore, traditional methods, like the f-x deconvolution method based on linear events assumption, are less effective in prestacked data. However, the basic assumption of the MDF is that seismic data usually have coherency among different spectral slices as well as the similarity between different patches. Thus, the MDF does not necessarily prefer a single domain (either stacked or prestacked volumes) as long as the similarity lies in the seismic data.

The first two key components of the MDF are to construct the multispectral data and normalize them, which are very fast to compute. The most time-consuming part is to use ITSReg to denoise the multispectral data, which contains a lot of tensor operations and also solves several optimization problems. The MDF, WNNM, and f-x deconvolution codes were run on MATLAB 2019a platform on an Intel Xeon Gold 6126 2.6 GHz processor. For a model size of 256×200 (size of synthetic-1 data), the average running time with ten tests for the MDF, WNNM, and f-x deconvolution method are 986.583, 100.159, and 0.112 s, respectively. For 2-D seismic data, we need to construct a higher dimensional data set (i.e., 3-D tensor) for the MDF, which introduces extra memory demand and computational cost. The memory demand and computational cost would be much larger for 3-D seismic data, which can be alleviated by parallel computing techniques in the future.

There is a possibility to extend this method to 3-D even 5-D processing. For the 3-D data set, the spectral input data set will be 4-D; thus, BM4D can be used to extract the similar patches among different spectral volumes, where the unfolding operator will unfold the tensor into a 3-D shape, which will be like the 2-D case in this article, then the problem can be solved likewise. For the 5-D data set, one way is to develop a higher dimensional block matching method to extract similar patches among different spectral volumes; the other way is to use a dimension reduction method and solve it in 3-D.

V. Conclusion

We present an MDF for seismic random noise attenuation. The proposed MDF includes three key components. First, we construct the multispectral seismic data by applying narrow-band filters on the noisy data. Second, we normalize the constructed multispectral data to mitigate the amplitude

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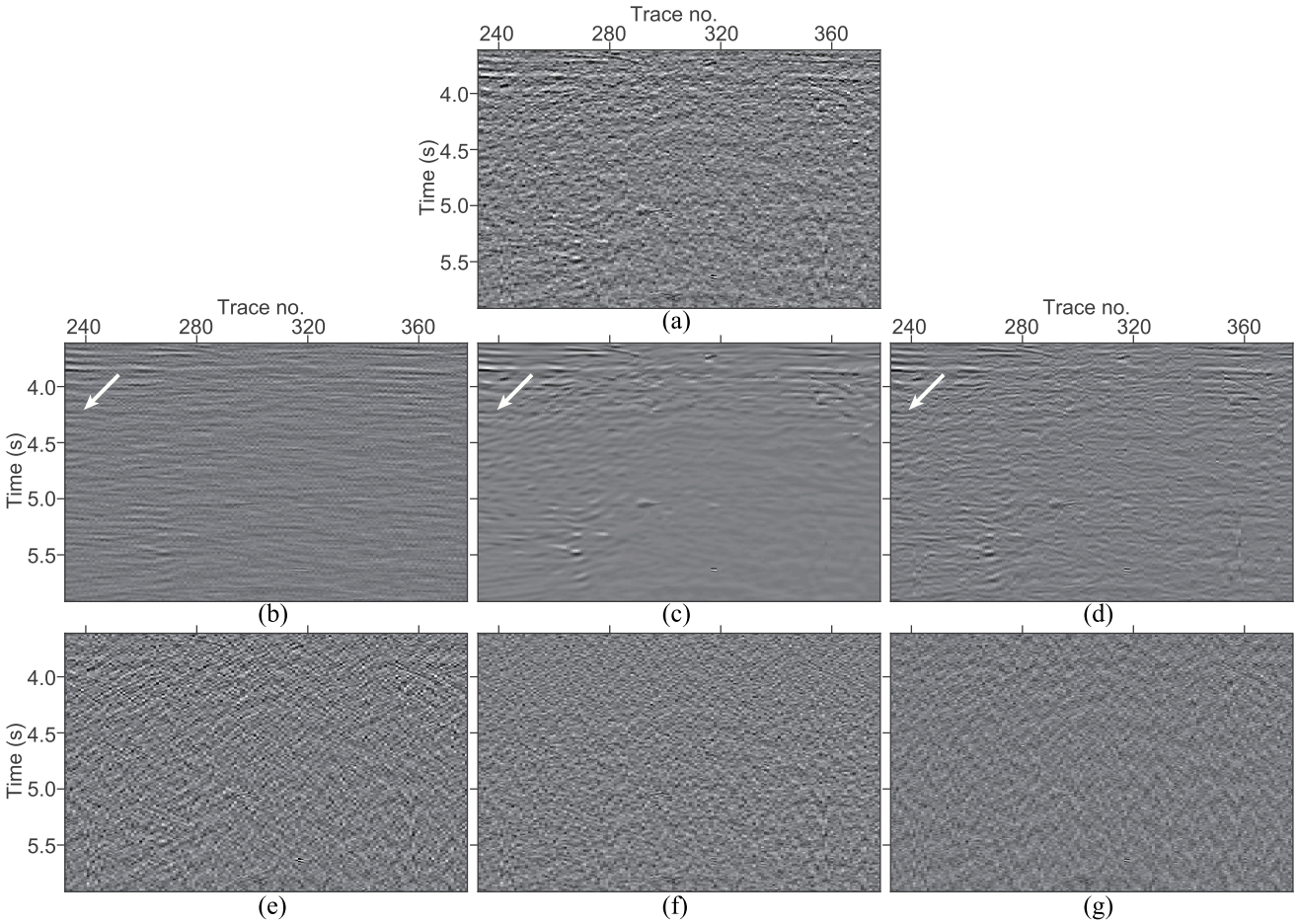


Fig. 21. Local details of the denoised results and corresponding noise sections for the white box area denoted by A in the field-1 data. (a) Enlarged view of the noisy data for the white box area denoted by A in Fig. 17. Denoised results are obtained by (b) f-x deconvolution method, (c) WNNM method, and (d) MDF, respectively. (e)–(g) Removed noise sections corresponding to (b)–(d), respectively.

discrepancy between different spectral slices. Third, we use the intrinsic tensor regularization method to denoise the multispectral data. Synthetic and field data examples show that the proposed method performs well both on prestack and stacked seismic data. It can well preserve the amplitude of useful signals while reducing random and erratic noises.

APPENDIX
INTRINSIC TENSOR SPARSITY REGULARIZATION

The third key component of the MDF is applying the ITSReg method [35] to suppress the random noise in the FBP. The ITSReg method was first proposed to denoise multispectral images and it combines the advantages of both Tucker [42] and CANDECOMP/PARAFAC decomposition [41], [50], [51] of a generic tensor; thus, it can provide a higher sparsity insight over the Tucker and CANDECOMP/PARAFAC decomposition. Here, we give a short review of the ITSReg method. The noisy and denoised FBP groups are denoted as $\mathcal{D}$ and $\mathcal{S}$, respectively. We can estimate $\mathcal{S}$ by solving the following optimization problem:

$$\mathcal{S} = \arg\min_{\mathcal{S}} \mathscr{H}(\mathcal{S}) + \frac{\gamma}{2}\|\mathcal{D} - \mathcal{S}\|_F^2 \tag{3}$$

where $\mathscr{H}(\mathcal{S})$ is the sparsity measure of $\mathcal{S}$ and γ is the hype-parameter which relates to the local noise variance of the data [35]. The Frobenius norm of a three-order tensor $\mathcal{T}$ has the form of $\|\mathcal{T}\|_F = (\sum_{i_1,i_2,i_3} t_{i_1 i_2 i_3}^2)^{1/2}$. Xie *et al.* [35] proposed to use the intrinsic tensor sparsity to measure the sparsity of $\mathcal{S}$ by

$$\mathscr{H}(\mathcal{S}) = \alpha\|\mathcal{C}\|_0 + (1-\alpha)\prod_{i=1}^{3}\operatorname{rank}(\mathbf{S}_{(i)}) \tag{4}$$

where $\mathcal{C}$ is the core tensor, which can be obtained by Tucker decomposition of the tensor $\mathcal{S}$, $\mathbf{S}_{(i)}$ is the mode-i unfolding of $\mathcal{S}$, α is a tradeoff parameter to compromise the contributions of these two terms and $\|\cdot\|_0$ describes the L_0 norm. Substituting (4) into (3), we have the following optimization problem:

$$\mathcal{S} = \arg\min_{\mathcal{S}} \alpha\|\mathcal{C}\|_0 + (1-\alpha)\prod_{i=1}^{3}\operatorname{rank}(\mathbf{S}_{(i)}) + \frac{\gamma}{2}\|\mathcal{D} - \mathcal{S}\|_F^2. \tag{5}$$

This optimization problem is NP-hard because the L_0 norm and rank terms in (5) can only take discrete values. Therefore, one can relax the intrinsic tensor sparsity as a log-sum form to

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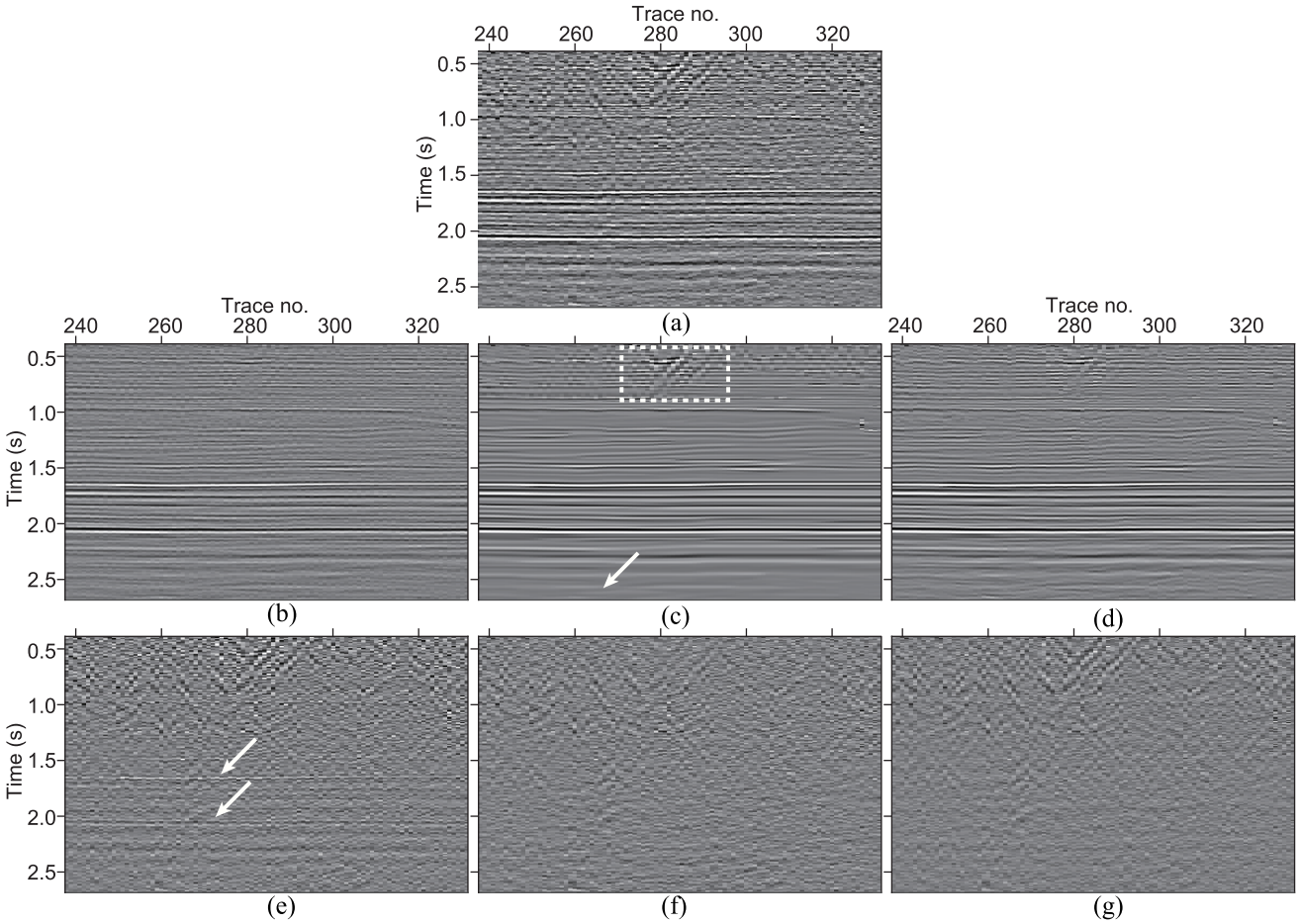


Fig. 22. Local details of the denoised results and corresponding noise sections for the white box area denoted by B in the field-1 data. (a) Enlarged view of the noisy data for the white box area denoted by B in Fig. 17. Denoised results are obtained by (b) f-x deconvolution method, (c) WNNM method, and (d) MDF, respectively. (e)–(g) Removed noise sections corresponding to (b)–(d), respectively.

reduce computation, which leads to the following optimization problem [35]:

$$\min_{\mathcal{S}} \mathscr{P}(\mathcal{C}) + \frac{1-\alpha}{\alpha} \prod_{j=1}^{3} \mathscr{P}^*(\mathbf{C}_{(j)}) + \frac{\gamma}{2\alpha} \|\mathcal{D} - \mathcal{S}\|_F^2 \tag{6}$$

where $\mathscr{P}(\mathcal{C}) = \sum_{i_1,i_2,i_3} \log(|c_{i_1 i_2 i_3}| + \varepsilon)$ and $\mathscr{P}^*(\mathbf{C}) = \sum_j \log[\sigma_j(\mathbf{C}) + \varepsilon]$, $\mathbf{C}_{(j)}$ is the mode-j matrix of $\mathcal{C}$, ε is a small positive number to make the expression in the log() bigger than 0, and $\sigma_j(\mathbf{C})$ denotes the jth singular value of $\mathbf{C}$. By introducing three auxiliary tensors $\mathcal{M}_j (j = 1, 2, 3)$, (6) can be equivalently rewritten as follows [35]:

$$\begin{aligned} \min_{\mathcal{C},\mathcal{M}_j,\mathbf{U}_j} & \mathscr{P}(\mathcal{C}) + \frac{1-\alpha}{\alpha} \prod_{j=1}^{3} \mathscr{P}^*(\mathbf{M}_{j(j)}) \\ & + \frac{\gamma}{2\alpha} \|\mathcal{D} - \mathcal{C} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_3\|_F^2 \\ \text{s.t. } & \mathcal{C} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_3 - \mathcal{M}_j = 0, \quad \mathbf{U}_j^T \mathbf{U}_j = \mathbf{I}, \quad j = 1, 2, 3 \end{aligned} \tag{7}$$

where $\mathbf{M}_{j(j)} = \text{unfold}_j(\mathcal{M}_j)$, $\mathbf{U}_j (j = 1, 2, 3)$ are the unitary matrices from the Tucker decomposition of $\mathcal{S}$. $\mathbf{I}$ is the identity matrix and $\text{unfold}_j(\mathcal{T})$ is the mode-j unfolding of a tensor $\mathcal{T}$. (7) can be solved by the alternating direction method of

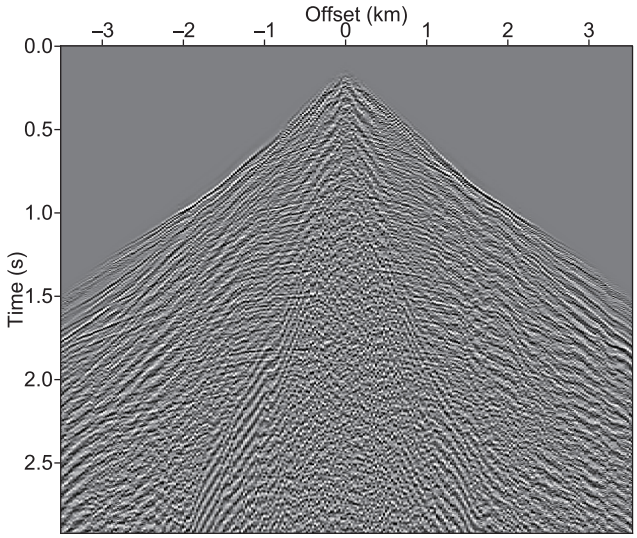


Fig. 23. Noisy field-2 data.

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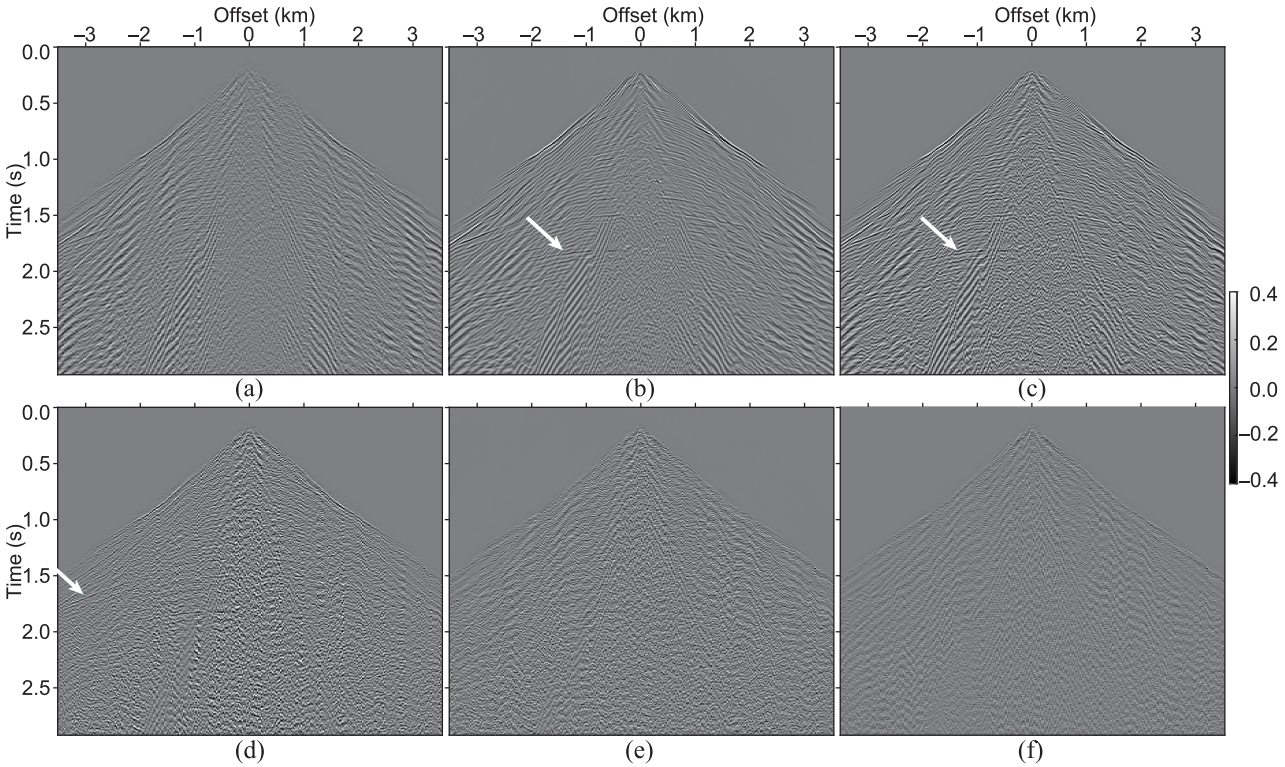


Fig. 24. Comparison of the denoised results and corresponding noise sections for the field-2 data. Denoised results are obtained by (a) f-x deconvolution method, (b) WNNM method, and (c) MDF, respectively. (d)–(f) Removed noise sections obtained by (a)–(c), respectively.

multipliers [52]–[54]. The augmented Lagrangian function of (7) is

$$\begin{aligned}\mathscr{L}_\mu(\mathcal{C},\mathcal{M}_1,\mathcal{M}_2,\mathcal{M}_3,\mathbf{U}_1,\mathbf{U}_2,\mathbf{U}_3)&=\mathscr{P}(\mathcal{C})+\frac{1-\alpha}{\alpha}\prod_{j=1}^{3}\mathscr{P}^*(\mathbf{M}_{j(j)})\\&+\frac{\gamma}{2\alpha}\|\mathcal{D}-\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3\|_F^2\\&+\sum_{j=1}^{3}\langle\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3-\mathcal{M}_j,\mathcal{G}_j\rangle\\&+\sum_{j=1}^{3}\frac{\mu}{2}\|\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3-\mathcal{M}_j\|_F^2\end{aligned}\tag{8}$$

where $\mathcal{G}_j$ are the Lagrange multipliers and μ is a positive constant.

With the other parameters fixed, $\mathcal{C}$ can be updated by solving $\min_{\mathcal{C}}\ \mathscr{L}_\mu(\mathcal{C},\mathcal{M}_1,\mathcal{M}_2,\mathcal{M}_3,\mathbf{U}_1,\mathbf{U}_2,\mathbf{U}_3)$, which is equivalent to the following subproblem:

$$\min_{\mathcal{C}}\ b\mathscr{P}(\mathcal{C})+\frac{1}{2}\|\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3-\mathcal{O}\|_F^2\tag{9}$$

where $b=1/(\gamma/\alpha+3\mu)$ and $\mathcal{O}=[(\gamma/\alpha)\mathcal{D}+\sum_j\mu\mathcal{M}_j-\mathcal{G}_j]/(\gamma/\alpha+3\mu)$. This subproblem has been proven to have a closed-form solution [35]

$$\mathcal{C}=\mathscr{T}_{b,\varepsilon}\left(\mathcal{O}\times_1\mathbf{U}_1^T\times_2\mathbf{U}_2^T\times_3\mathbf{U}_3^T\right)\tag{10}$$

which is defined as

$$\mathscr{T}_{b,\varepsilon}(x)=\begin{cases}0, & \text{if } c_2\le 0\\ \operatorname{sign}(x)\left(\dfrac{c_1+\sqrt{c_2}}{2}\right), & \text{if } c_2>0\end{cases}\tag{11}$$

where $c_1=|x|-\varepsilon$ and $c_2=(c_1)^2-4(b-\varepsilon|x|)$.

When updating $\mathbf{U}_k(k=1,2,3)$ with the other parameters fixed, one can also deduce its closed-form solution, which is equivalent to the following problem:

$$\min_{\mathbf{U}_k}\ \|\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3-\mathcal{O}\|_F^2\tag{12}$$

and its closed-form solution is

$$\mathbf{U}_k=\mathbf{A}_k\mathbf{B}_k^T\tag{13}$$

where $\mathbf{A}_k\sum\mathbf{B}_k^T$ is the SVD decomposition of $\mathbf{E}_k$, and $\mathbf{E}_k$ is

$$\begin{bmatrix}\mathbf{E}_1\\ \mathbf{E}_2\\ \mathbf{E}_3\end{bmatrix}=\begin{bmatrix}\mathbf{O}_{(1)}\text{unfold}_1(\mathcal{C}\times_2\mathbf{U}_2\times_3\mathbf{U}_3)\\ \mathbf{O}_{(2)}\text{unfold}_2(\mathcal{C}\times_1\mathbf{U}_1\times_3\mathbf{U}_3)\\ \mathbf{O}_{(3)}\text{unfold}_3(\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2)\end{bmatrix}\tag{14}$$

where $\mathbf{O}_{(k)}$ is the mode-k unfolding of the tensor $\mathcal{O}$.

Additionally, with $\mathcal{M}_j(j\ne k,j,k\in[1,2,3])$ and other parameters fixed, $\mathcal{M}_k$ can be updated by solving the following subproblem:

$$\min_{\mathcal{M}_k}\ q_k\mathscr{P}^*(\mathbf{M}_{k(k)})+\frac{1}{2}\left\|\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3+\frac{1}{\mu}\mathcal{G}_k-\mathcal{M}_k\right\|_F^2\tag{15}$$

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where $q_k = (1-\alpha)/(\alpha\mu)\prod_{j\neq k}\mathscr{P}^*(\mathbf{M}_{j(j)})$, which has a closed-form solution

$$\mathcal{M}_k = \text{fold}_k\left(\mathbf{V}_1\sum\nolimits_{q_k}\mathbf{V}_2^T\right) \tag{16}$$

where $\sum_{q_k} = \text{diag}[\mathscr{T}_{q_k,\varepsilon}(\sigma_1), \mathscr{T}_{q_k,\varepsilon}(\sigma_2), \ldots, \mathscr{T}_{q_k,\varepsilon}(\sigma_n)]$ and $\mathbf{V}_1\text{diag}(\sigma_1,\sigma_2,\ldots,\sigma_n)\mathbf{V}_2^T$ is the SVD decomposition of $\text{unfold}_k(\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3+\mathcal{G}_k/\mu)$. Algorithm 2 shows the implementation details of the ITSReg method (refer to [35] for more details).

Algorithm 2 $\mathcal{S} = \text{ITSReg}(\mathcal{D})$

Input: Noisy FBP group $\mathcal{D}$, constant parameter ρ
Output: Denoised FBP group $\mathcal{S}$

while not convergence **do**
  Update $\mathcal{C}$ by (10)
  Update $\mathbf{U}_k$ by (13), $\forall k = 1, 2, 3$
  Update $\mathcal{M}_k$ by (16), $\forall k = 1, 2, 3$
  Update the multipliers and let $\mu := \rho\mu$
end while
Calculate $\mathcal{S} = \mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3$
return $\mathcal{S}$

ACKNOWLEDGMENT

The authors would like to thank three anonymous reviewers for their valuable comments and helpful suggestions, which greatly improve the quality of the paper.

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